KAIZEN MATHS

Lesson starter

Basic Differentiation of Polynomials

Gradients, Tangents and Normals

Name

S1

Differentiate with respect to x.

$f(x) = 6x^2 + 5x + 9$

S2

Differentiate with respect to x.

$f(x) = 5x^2 + 6x + 7,$ find the tangent at $x = 3$

S3

Differentiate with respect to x.

$f(x) = 3x^2 + 2x + 5,$ find the normal at $x = 2$

S4

Differentiate with respect to x.

$f(x) = 6x^2 + 5x + 3$

Show your working

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ALG-E739-L3 · S

Show your working

KAIZEN MATHS

Exit ticket

Basic Differentiation of Polynomials

Gradients, Tangents and Normals

Name _____

E1

Differentiate with respect to x.

$f(x) = 2x^2 + 3x + 3,$ find the tangent at $x = 4$

E2

Differentiate with respect to x.

$f(x) = 2x^2 + 6x + 7,$ find the normal at $x = 2$

E3

Differentiate with respect to x.

$f(x) = 4x^2 + 2x + 3$

Show your working

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Exit ticket

Basic Differentiation of Polynomials

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Differentiate with respect to x.

$f(x) = 2x^2 + 3x + 3,$ find the tangent at $x = 4$

E2

Differentiate with respect to x.

$f(x) = 2x^2 + 6x + 7,$ find the normal at $x = 2$

E3

Differentiate with respect to x.

$f(x) = 4x^2 + 2x + 3$

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Show your working

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Exit ticket

Basic Differentiation of Polynomials

Gradients, Tangents and Normals

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E1

Differentiate with respect to x.

$f(x) = 2x^2 + 3x + 3,$ find the tangent at $x = 4$

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Differentiate with respect to x.

$f(x) = 2x^2 + 6x + 7,$ find the normal at $x = 2$

E3

Differentiate with respect to x.

$f(x) = 4x^2 + 2x + 3$

ALG-E739-L3 · E

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$f(x) = 2x^2 + 6x + 7,$ find the normal at $x = 2$

E3

Differentiate with respect to x.

$f(x) = 4x^2 + 2x + 3$

ALG-E739-L3 · E

Show your working

Teacher answers & working

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S1

Differentiate with respect to x.

$f(x) = 6x^2 + 5x + 9$

Answer:

$f'(x) = 12x + 5$

Working:

Use the power rule.

Substitute into the power rule.

Multiply the coefficient; subtract 1 from the power.

Write this term's derivative.

Substitute into the power rule.

Multiply the coefficient; subtract 1 from the power.

Write this term's derivative.

Differentiate the constant.

Combine the term derivatives.

Substitute the given x-value into the derivative.

State the gradient.

Teacher answers & working

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S2

Differentiate with respect to x.

$f(x) = 5x^2 + 6x + 7,\ \text{find the tangent at } x = 3$

Answer:

Tangent: $y = 36x - 38$

Working:

Use the power rule before substituting any values.

Substitute into the power rule.

Multiply the coefficient; subtract 1 from the power.

Write this term's derivative.

Substitute into the power rule.

Multiply the coefficient; subtract 1 from the power.

Write this term's derivative.

Differentiate the constant.

Combine the term derivatives.

Find the point on the curve using the given x-value.

Find the tangent gradient using the same x-value.

Use point-gradient form.

Substitute the point and gradient.

Expand the bracket.

Add 70 to both sides.

Simplify.

Teacher answers & working

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S3

Differentiate with respect to x.

$f(x) = 3x^2 + 2x + 5,$ **find the normal at** $x = 2$

Answer:

Normal: $y = -\frac{1}{14}x + 21.14$

Working:

Use the power rule before substituting any values.

Substitute into the power rule.

Multiply the coefficient; subtract 1 from the power.

Write this term's derivative.

Substitute into the power rule.

Multiply the coefficient; subtract 1 from the power.

Write this term's derivative.

Differentiate the constant.

Combine the term derivatives.

Find the point on the curve using the given x-value.

Find the tangent gradient first.

The normal gradient is the negative reciprocal of the tangent gradient.

Use point-gradient form.

Substitute the point and normal gradient.

Expand the bracket.

Add 21 to both sides.

Simplify the intercept (round if needed).

Teacher answers & working

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S4

Differentiate with respect to x.

$f(x) = 6x^2 + 5x + 3$

Answer:

$f'(x) = 12x + 5$

Working:

Use the power rule.

Substitute into the power rule.

Multiply the coefficient; subtract 1 from the power.

Write this term's derivative.

Substitute into the power rule.

Multiply the coefficient; subtract 1 from the power.

Write this term's derivative.

Differentiate the constant.

Combine the term derivatives.

Substitute the given x-value into the derivative.

State the gradient.

Teacher answers & working

Basic Differentiation of Polynomials · Gradients, Tangents and Normals

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E1

Differentiate with respect to x.

$f(x) = 2x^2 + 3x + 3,$ **find the tangent at** $x = 4$

Answer:

Tangent: $y = 19x - 29$

Working:

Use the power rule before substituting any values.

Substitute into the power rule.

Multiply the coefficient; subtract 1 from the power.

Write this term's derivative.

Substitute into the power rule.

Multiply the coefficient; subtract 1 from the power.

Write this term's derivative.

Differentiate the constant.

Combine the term derivatives.

Find the point on the curve using the given x-value.

Find the tangent gradient using the same x-value.

Use point-gradient form.

Substitute the point and gradient.

Expand the bracket.

Add 47 to both sides.

Simplify.

Teacher answers & working

Basic Differentiation of Polynomials · Gradients, Tangents and Normals

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E2

Differentiate with respect to x.

$f(x) = 2x^2 + 6x + 7,$ **find the normal at** $x = 2$

Answer:

Normal: $y = -\frac{1}{14}x + 27.14$

Working:

Use the power rule before substituting any values.

Substitute into the power rule.

Multiply the coefficient; subtract 1 from the power.

Write this term's derivative.

Substitute into the power rule.

Multiply the coefficient; subtract 1 from the power.

Write this term's derivative.

Differentiate the constant.

Combine the term derivatives.

Find the point on the curve using the given x-value.

Find the tangent gradient first.

The normal gradient is the negative reciprocal of the tangent gradient.

Use point-gradient form.

Substitute the point and normal gradient.

Expand the bracket.

Add 27 to both sides.

Simplify the intercept (round if needed).

Teacher answers & working

Basic Differentiation of Polynomials · Gradients, Tangents and Normals

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E3

Differentiate with respect to x.

$f(x) = 4x^2 + 2x + 3$

Answer:

$f'(x) = 8x + 2$

Working:

Use the power rule.

Substitute into the power rule.

Multiply the coefficient; subtract 1 from the power.

Write this term's derivative.

Substitute into the power rule.

Multiply the coefficient; subtract 1 from the power.

Write this term's derivative.

Differentiate the constant.

Combine the term derivatives.

Substitute the given x-value into the derivative.

State the gradient.